TECHNICAL REPORT · FTALGO 0.3

Factoring and chemistry on a simulated fault-tolerant quantum computer

Algorithms run end to end on a surface-code machine whose logical error rates were measured by circuit-level simulation

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the explainer site · github.com/Jaspersands/ft-algorithms ·

SUMMARY

Shor's algorithm (full modular exponentiation, with textbook ripple-carry arithmetic and with windowed arithmetic using measurement-based uncomputation; semiclassical approximate QFT) and iterative phase estimation of molecular ground-state energies (H_2, HeH^+ and a small LiH active space) are run end to end on a simulated rotated-surface-code machine. Every logical error channel, idling, lattice-surgery merges and the surgery CNOT, was measured in 316 circuit-level experiments (0.82×10^9 shots) with stabilizer-qec under SD6 noise and extracted with baselines propagated through each operation; magic states come from published factory models. Whole surgery experiments predicted from their parts agree with circuit-level measurements to 0.93–1.26× (median 1.04×). Windowed arithmetic cuts Toffolis by 2.6–3.0× and rounds by 1.4–1.8× on the simulated moduli. Extrapolated with the same error model to RSA-2048, textbook arithmetic needs d = 35, 32.5 M physical qubits and 1.96 years; windowed arithmetic 298 days with 43.2 M. At their authors' own failure budgets, the model gives Lee et al.'s FeMoco estimate its published d = 31, and Gidney's 2025 RSA-2048 estimate d = 29 rather than 25. Memories under three more circuit-level models put the distance for 10^{-15} per patch-round between 23 and 29, a range that contains his choice and of which the calibration's circuit, with separate measurement and reset layers, is the pessimistic end.

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1 Introduction

Resource estimates for fault-tolerant algorithms usually multiply a gate count by an assumed logical error rate. This work runs the algorithms instead. A logical-level simulator (ftsim, written for this project) executes the actual circuits, with every operation followed by its measured logical channel and every waiting patch accumulating measured idle error. The results are success probabilities as functions of code distance, physical error rate and magic-state source, with error budgets that say which operations cause the failures.

The approach rests on one fact and one approximation. The fact: under Pauli circuit noise, for a Clifford circuit decoded into a Pauli-frame correction, the logical action of one surface-code operation is exactly a Pauli channel, since every syndrome history ends in some logical Pauli. The approximation: consecutive operations' channels are composed as independent. Section 3.2 measures how good that is.

Contributions:

- A calibrated logical model of a rotated surface code under SD6 noise: idle (including logical Y), Z⊗Z and X⊗X merges and the surgery CNOT, from 316 experiments, with fits in d and p.
- A composition test: whole lattice-surgery experiments predicted from parts at the logical level and compared with circuit-level measurement.
- Shor's algorithm with honest arithmetic (no use of the period anywhere in the circuit) run on that model for N up to 143, scored by peak probability because, for small N, uniformly random outcomes already "factor" with probability 1.00 (N = 15); and the same with windowed arithmetic and measurement-based uncomputation.
- Phase estimation of molecular ground-state energies to chemical accuracy, from Hamiltonians computed from scratch (H_2, HeH^+) and from a frozen-core CASSCF active space (LiH).
- Extrapolation to RSA-2048 and FeMoco, and a check of two published estimates' code distances against the measured model at their own failure budgets.

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2 Methods

2.1 Circuit-level calibration

All physical-level simulation uses stabilizer-qec 1.2 (the author's engine, which matches Stim and PyMatching) with SD6 noise at p ∈ {0.1, 0.2, 0.3, 0.5}% and correlated matching. Experiments: memories of d, 2d, 3d and 4d rounds in both bases (d = 3 … 11), reference-qubit memories (d ≤ 7), Z⊗Z and X⊗X merges of d rounds with d rounds apart before and after, the surgery CNOT (two merges and an ancilla patch, 4d rounds in all), and, for the composition test, three Z⊗Z measurements in a row and a three-patch merge. Each ran to 500 failures or a 300-second cap; every record keeps its seed.

Channels through fidelities. A distribution over flip patterns has Pauli fidelities f(s) = Σ_x P(x)(−1)^(s·x), which multiply under composition. The per-round idle channel is the slope of log f over the memory lengths (preparation and readout cancel). An operation's own channel is the measured one divided by its baseline, where the baseline (preparation plus d rounds before, d rounds plus readout after, each the square root of a 2d-round memory's fidelity) is first pushed through the ideal operation: a CNOT copies an X error on the control onto the target and a Z on the target onto the control, and an error before a merge changes its outcome. Dividing naively gives negative probabilities for "X on the control only". Uncertainties are bootstrap (300 resamples).

Logical Y. X- and Z-basis memories see X∪Y and Z∪Y. A reference-qubit experiment gives the full channel: the patch is projected noiselessly (every Z check by a noiseless Pauli-product measurement, which round 1 is compared with), entangled with a perfect qubit by a noiseless Z_R Z_L measurement, run, and ended by a perfect round and noiseless X_R X_L and Z_R Z_L measurements. Without the initial projection an X error before the first Z-check readout flips Z_R Z_L unseen.

Fits. Each component ε(d, p) is fitted per p as log ε = a + b(d + 1)/2 by weighted least squares over points with at least ten events, and globally as A(p/p*)^((d+1)/2). Values beyond the largest measured distance are extrapolations and are marked as such in every figure.

2.2 The machine

operation	rounds	logical channel
X, Y, Z	0	none (Pauli frame)
H	0	none: transversal, orientation tracked in the fast block †
S, S†	d	Z⊗Z merge channel (
T, T†	d	merge channel + Z w.p. ε_T † + wrong outcome → Z w.p. ½ (twirled S)
CX, CZ	2d	measured surgery-CNOT channel, idling included
CCX, CCZ	2d	3 merges + Z-type w.p. ε_CCZ † + wrong outcomes → twirled CZ; d rounds idle
M, R	1	one round of idling
idle	per round	measured (p_X, p_Y, p_Z)
feed-forward	10	decoder reaction time before a TABLE †

† marks cited inputs. Magic states: injection ($\varepsilon_T = p$, conservative against Li 2015's $\approx 0.4p$), 15-to-1 distillation (Litinski 2019, Table 1: 4.5×10^{-8} per T and 5.2×10^{-11} per CCZ at $p = 0.1\%$) and cultivation (Gidney, Shutty and Jones 2024: 2×10^{-9} at $p = 0.1\%$, 4×10^{-11} at 0.05%; $|\mathrm{CCZ}\rangle$ by 8T-to-CCZ, $28\,\varepsilon_T^2$). Physical qubits: Litinski's fast block, $2n + \lceil\sqrt{8n}\rceil + 1$ tiles of $2(d+1)^2$ qubits, plus factories sized by their published qubit·rounds per state. One round is 1 µs.

The noise compiler gives every qubit a clock. An operation starts when its qubits are free (a TABLE also waits for its records plus the reaction time); each waiting qubit receives the composed idle channel for its gap; the operation's channel follows it with a class label for attribution. REPEAT blocks are barriers, scheduled once; TABLE cases are padded to the longest, so the schedule does not depend on outcomes. Channel composition is written with log1p/expm1: per-round rates reach 10^{-20} at large d and a naive $1 - (1 - 2q)^n$ cancels to zero.

2.3 The simulator

ftsim (Rust) runs programs in a Stim-like format: Clifford + T + Toffoli gates, Pauli channels on one to three qubits, measurements, REPEAT, and TABLE (classical feed-forward choosing a block by earlier records). Backends: a sparse state (hash map of basis states; runs of permutation and phase gates fused per basis state), which handles 36-qubit Shor circuits in milliseconds per shot, and a dense vector. Each shot's RNG derives from (seed, shot), so results do not depend on threads.

Stratified sampling. With K the number of static faults, $P(\text{success}) = \Sigma_k\, P(K = k)\cdot S_k$. $P(K = k)$ is computed exactly (a truncated generating polynomial over the site tree, REPEAT by exponentiation), and S_k is estimated from shots with exactly k faults, drawn by sampling k sites with weights $q/(1-q)$ and rejecting duplicates, which is exactly the conditional law. Strata beyond the tail cutoff carry their mass as an interval. The one-fault stratum, with faults logged, gives each operation class's harm.

2.4 Algorithms

Textbook Shor. Cuccaro ripple-carry adders; modular addition of a classical constant with a sign flag (Vedral–Barenco–Ekert, Beauregard); controlled modular multiplication with uncomputation by the inverse and a controlled swap; $3n + 6$ qubits and $2n(2n(10n + 12) + n)$ Toffolis. One control qubit is recycled through $2n$ rounds of a semiclassical approximate QFT whose correction rotations, chosen by TABLE from the previous $\lceil \log_2 2n \rceil + 2$ records, are synthesized by gridsynth to $\varepsilon = 10^{-3}/2n$ and verified as matrices. The base is $a = 2$ for every N. Every multiplier is the generic circuit, even when its constant is 1.

Windowed Shor with measurement-based uncomputation. The adder computes each carry with a Toffoli into a fresh qubit and, instead of uncomputing it with a second Toffoli, measures it in the X basis and applies CZ to the carry's two inputs when the outcome is 1 (Gidney 2018), a TABLE in the program; subtraction is addition under bitwise complement. The exponent is read two bits at a time: for each window the multiple $a^{(2^j \cdot m)} \bmod N$ ($m \in \{1, 2, 3\}$) is looked up into an $(n + 1)$-qubit register and added under modular reduction, halving the number of controlled multiplications. The circuit has $4n + 7$ qubits; its Toffoli count is a closed form equal to the compiled circuits' for every simulated N.

Chemistry. STO-3G integrals of s Gaussians in closed form (Boys F_0), restricted Hartree–Fock with DIIS, Jordan–Wigner with interleaved spins, and Z_2 tapering of both spin parities (Bravyi et al.). Iterative phase estimation reads 10 bits of $U = \exp(-i(H - E_{HF})\tau)$ with $\tau = 2\pi$, approximated by four fourth-order Suzuki steps; controlled Pauli rotations use only uncontrolled synthesized Rz's, so their global phases stay global.

LiH active space. LiH in STO-3G has six spatial orbitals. Integrals come from PySCF; the Li $1s^2$ core is frozen into an effective one-body term, $h_{pq} + \Sigma_c\, [2(pq|cc) - (pc|cq)]$, and the two valence electrons are placed in two orbitals optimised by CASSCF(2, 2), which after tapering leaves a 2-qubit, 9-term Hamiltonian. Its ground energy equals PySCF's CASSCF energy to 10^{-9} Ha. It is a deliberately small problem: its value here is a third molecule through the same pipeline, not chemical scale.

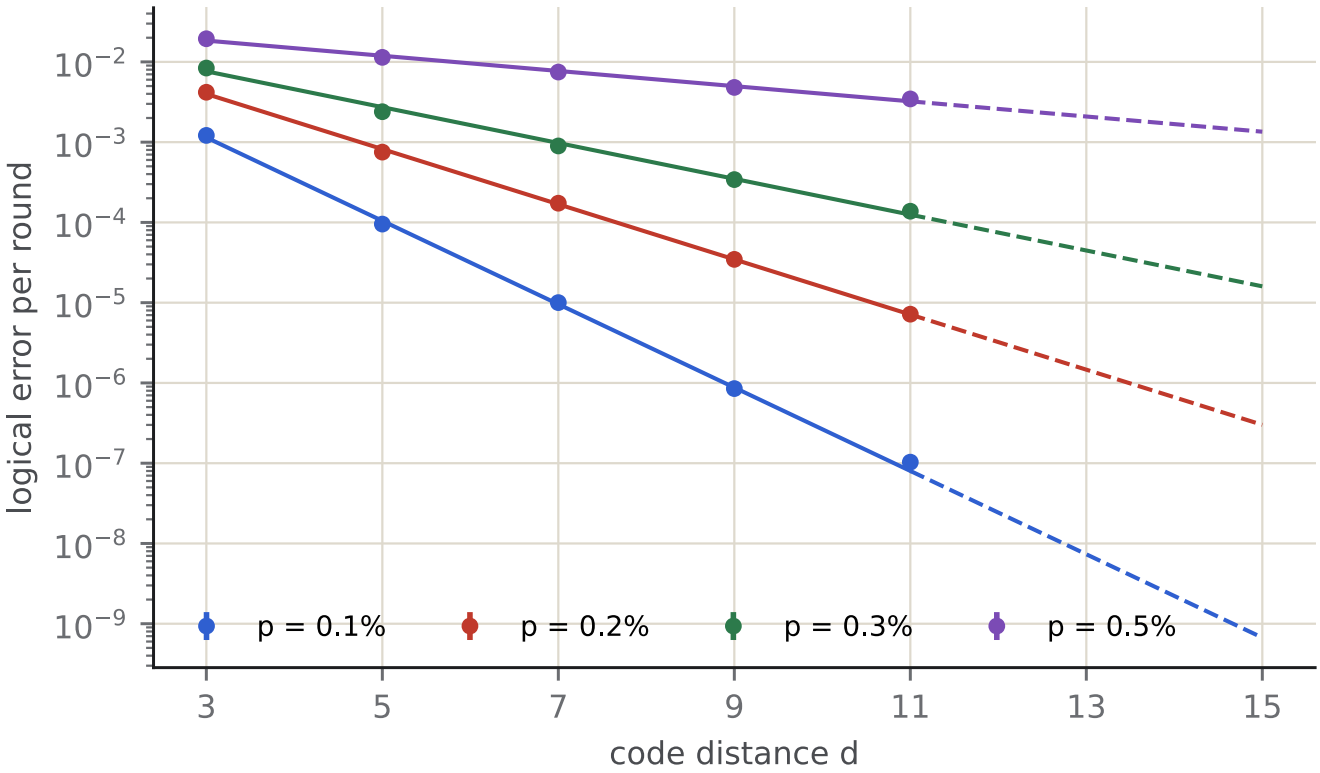
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3 Results

3.1 The calibrated machine

316 experiments, 0.82×10^9 shots and 459,686 logical failures. Suppression factors Λ (per two units of distance):

p	idle (XuY)	idle (ZuY)	CNOT X on target	CNOT Z on control	Z⊗Z patch Z	Z⊗Z wrong outcome
0.1%	10.93	10.06	7.91	7.10	7.12	12.30
0.2%	4.86	4.72	3.42	3.43	3.17	6.33
0.3%	2.80	2.85	2.01	2.01	1.98	2.97
0.5%	1.55	1.62	1.13	1.11	1.11	1.58

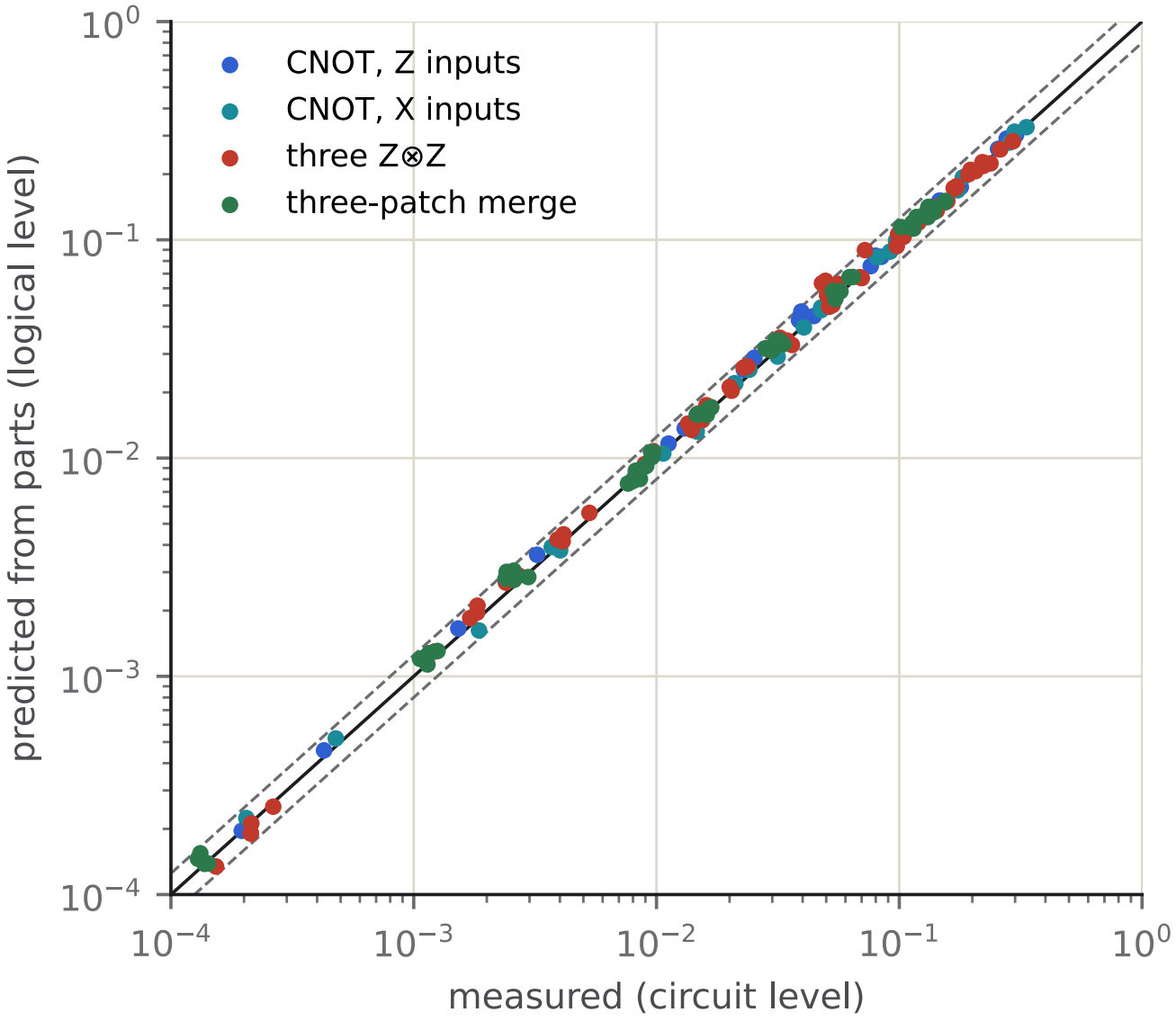


Per-round logical error of an idle patch (X∪Y). Points measured (1σ), lines per-p fits, dashed beyond the largest measured distance.

At p = 0.1% an idle patch errs 8.8×10^{-7} per round at d = 9 (X∪Y), and the fit gives 4.3×10^{-15} at d = 25 for X∪Y alone (1.3×10^{-14} for any Pauli; Section 3.8). Logical Y is 4% of the X∪Y rate at d = 3, 43× what independent X and Z would give: Y errors are included as measured.

Cross-check. 36 experiments rerun with Stim and PyMatching (correlated) agree with stabilizer-qec within statistics (largest $|z| = 2.2$).

3.2 Composition



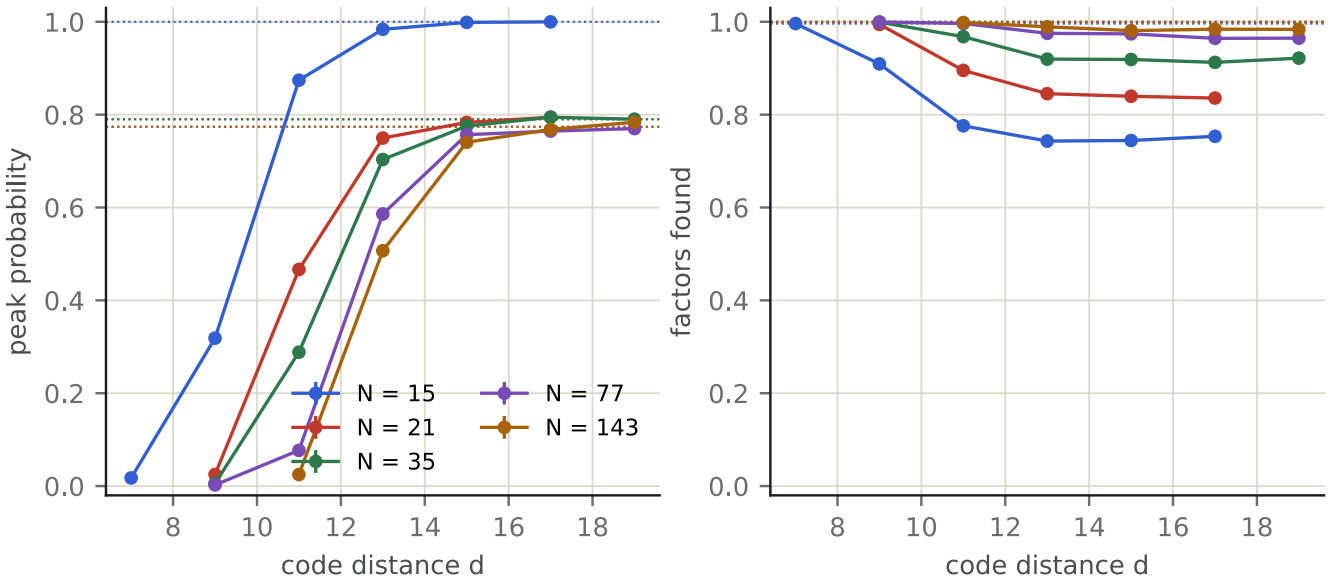
Observable failure rates of whole surgery experiments: predicted from calibrated parts by ftsim vs measured at circuit level; dashed ±25%.

Across 48 experiment–noise–distance combinations the predicted failure rate is 0.93–1.26× the measured one, median 1.04×. The independent-composition model is accurate to about 10% and leans pessimistic.

3.3 Shor's algorithm

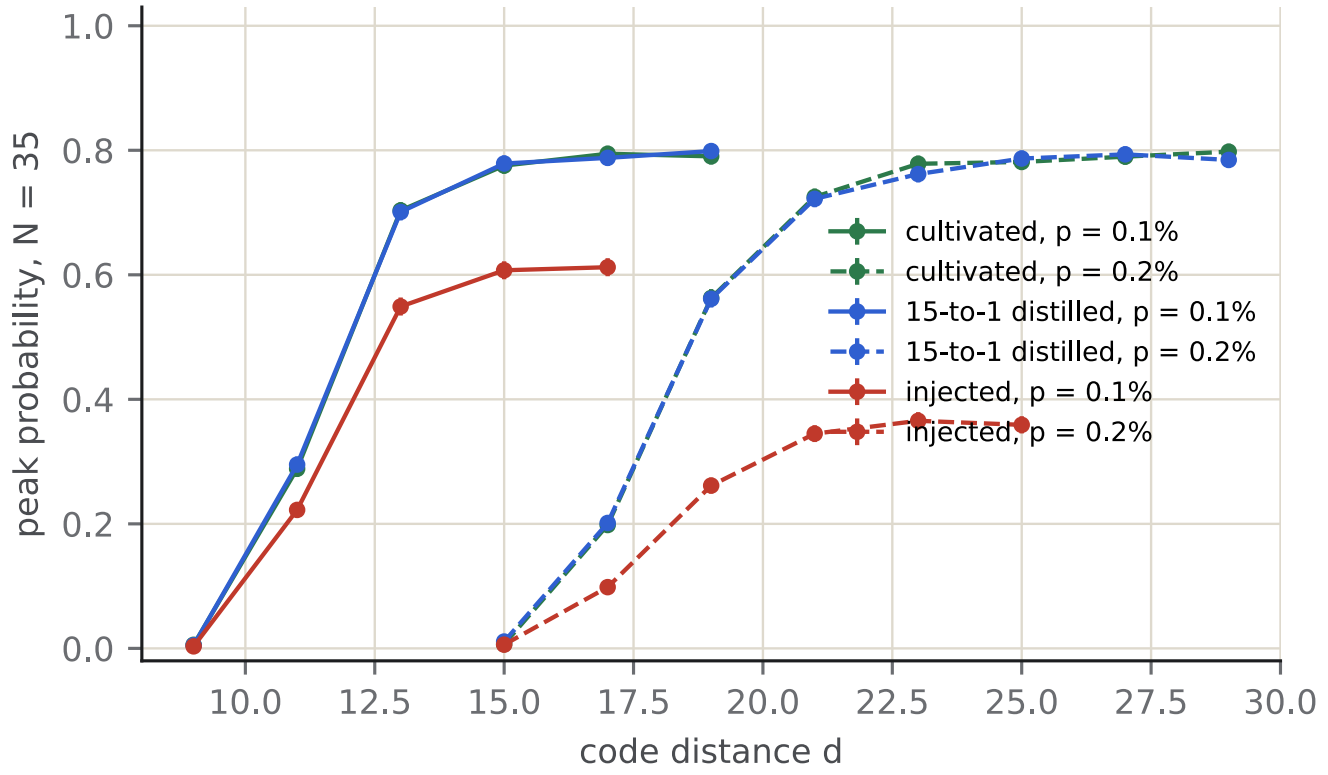
At p = 0.1% with cultivated magic states:

N	n	logical qubits	Toffolis	T	noiseless peak	random peak	random factors	d for 90%	physical qubits	run time
15	4	18	3,360	223	1.000	0.0156	0.996	13	55.4 k	225 ms
21	5	21	6,250	317	0.790	0.0059	0.999	13	58.1 k	418 ms
35	6	24	10,440	425	0.790	0.0029	1.000	15	63.4 k	808 ms
77	7	27	16,170	519	0.774	0.0018	1.000	15	67 k	1.25 s
143	8	30	23,680	635	0.774	0.0009	1.000	15	70.8 k	1.81 s



Left: peak probability vs d (cultivated states, p = 0.1%); dotted: noiseless. Right: the same runs scored by factors found; dotted: random-outcome baseline.

Factoring success is a poor score for small N. Classical post-processing that tries the convergents' small multiples finds the factors from most outcomes, and the noisiest runs look best. The peak probability falls to its random baseline (r/2^m) instead.



N = 35: peak probability vs d for each magic-state source; solid p = 0.1%, dashed p = 0.2%.

magic states	p	ε_T †	ε_CCZ †	best peak	d for 90%
cultivated	0.1%	2.0×10^{-9}	1.1×10^{-16}	0.794 (d = 17)	15
cultivated	0.2%	1.0×10^{-7}	2.8×10^{-13}	0.798 (d = 29)	21
15-to-1 distilled	0.1%	4.5×10^{-8}	5.2×10^{-11}	0.799 (d = 19)	15
15-to-1 distilled	0.2%	3.6×10^{-7}	3.3×10^{-9}	0.793 (d = 27)	21
injected	0.1%	1.0×10^{-3}	2.8×10^{-5}	0.612 (d = 17)	never
injected	0.2%	2.0×10^{-3}	1.1×10^{-4}	0.366 (d = 23)	never

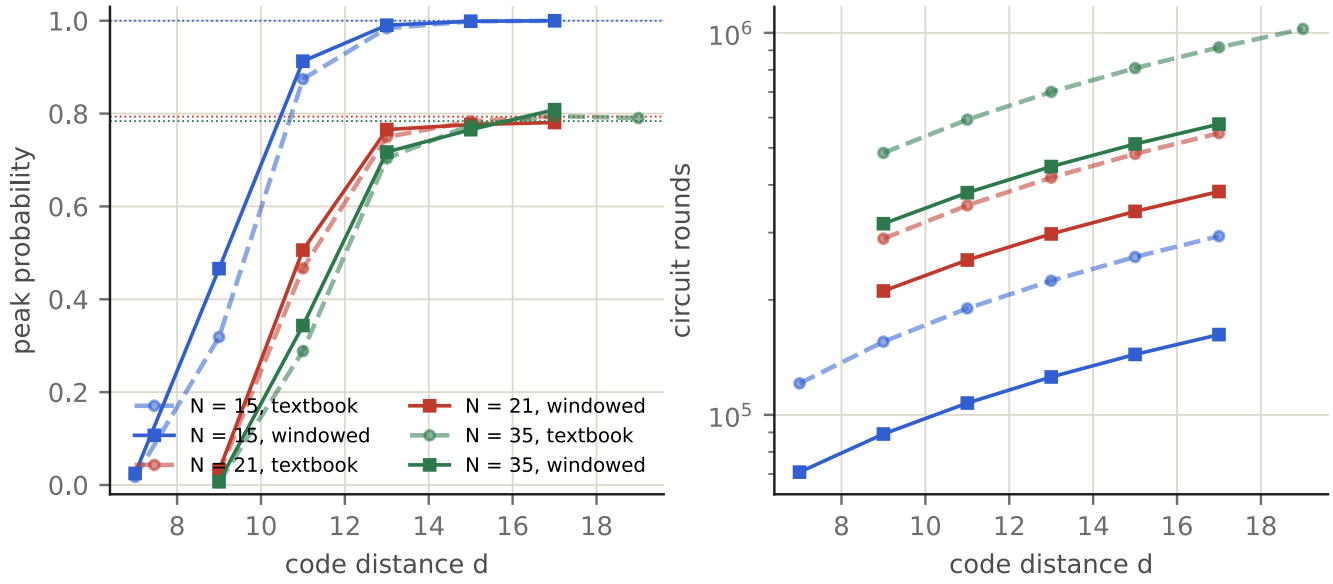
Error budget (N = 35, d = 15, cultivated, p = 0.1%):

class	expected faults	share	score of one-fault runs
idle	0.027	80.3%	0.394
CNOT	5.3×10^{-3}	15.9%	0.564
CCZ	1.2×10^{-3}	3.7%	0.667
T	1.7×10^{-5}	0.0%	–
S	1.0×10^{-5}	0.0%	–
measure	3.7×10^{-8}	0.0%	–

Idling accounts for 80% of the expected faults. The textbook circuit runs one Toffoli at a time (the schedule averages 5.13·d rounds per Toffoli with 3.2 patches busy) while every other patch waits. The last column is the peak probability of runs with exactly one fault of that class: a single logical fault is far from always fatal.

3.4 Windowed arithmetic

N	arithmetic	logical qubits	Toffolis	rounds (d=11)	peak (d=11)	d for 90%	physical qubits	run time
15	textbook (Cuccaro)	18	3,360	189,969	0.874	13	55.4 k	225 ms
15	windowed (k = 2, MBU)	23	1,248	107,387	0.913	11	46.1 k	107 ms
21	textbook (Cuccaro)	21	6,250	353,564	0.467	13	58.1 k	418 ms
21	windowed (k = 2, MBU)	27	2,390	254,313	0.506	13	47 k	298 ms
35	textbook (Cuccaro)	24	10,440	592,744	0.288	15	63.4 k	808 ms
35	windowed (k = 2, MBU)	31	3,528	381,614	0.343	13	50.2 k	447 ms

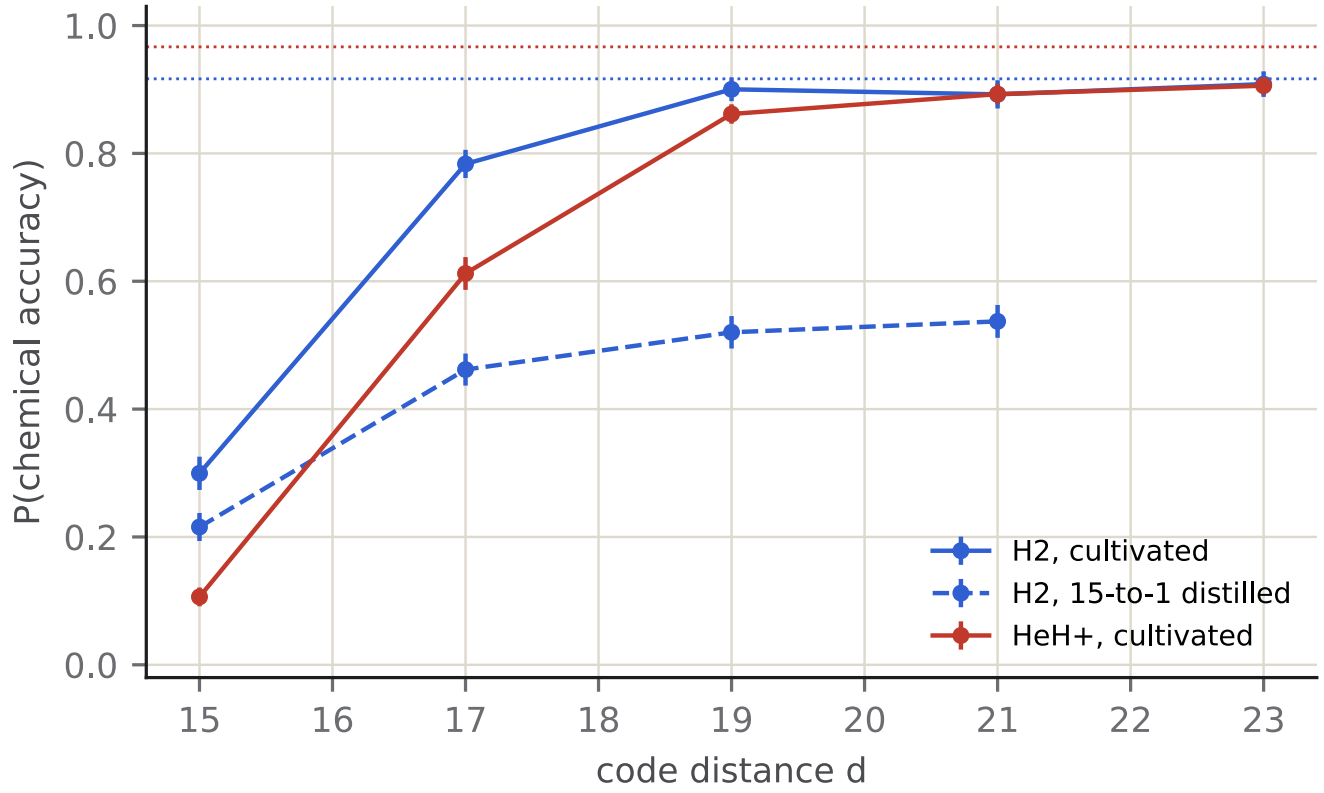


Windowed (solid) vs textbook (dashed) Shor on the simulated machine (cultivated magic states, p = 0.1%). Left: peak probability vs d. Right: rounds.

On N = 15, 21, 35 the windowed circuit uses 2.6–3.0× fewer Toffolis and 1.4–1.8× fewer rounds at d = 11, for 5–7 more logical qubits. Rounds fall less than Toffolis because each measurement-based uncomputation waits for the decoder before its CZ correction. Shorter runs idle less: the distance for 90% of the noiseless peak is N = 15: 13 → 11; N = 21: 13 → 13; N = 35: 15 → 13.

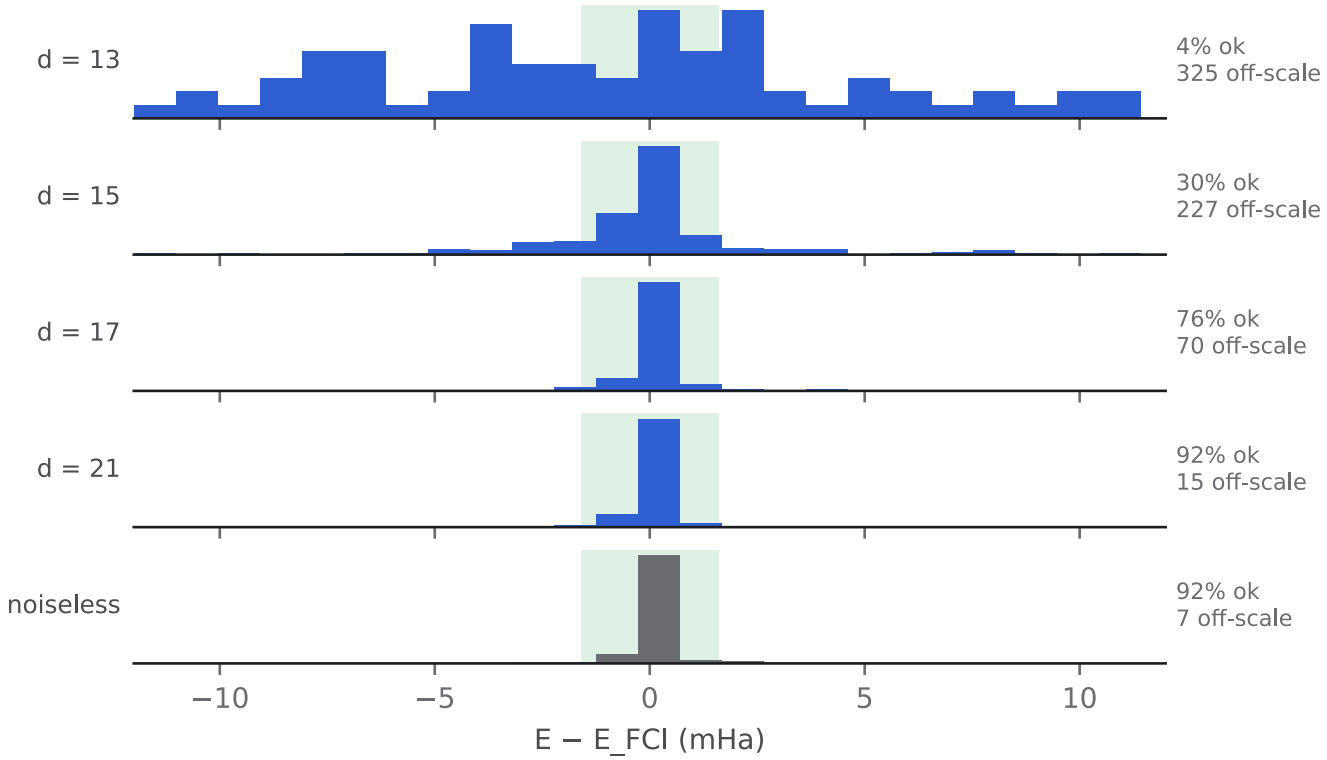
3.5 Phase estimation

molecule	logical qubits	terms	T gates	E_HF (Ha)	E_FCI (Ha)	noiseless	best	d for 90%	run time	physical qubits
H2	3	5	1.8×10^{7}	-1.116714	-1.137276	0.92	0.91 (d = 23)	19	7.2 min	22.1 k
HeH+	3	9	4.4×10^{7}	-2.841836	-2.851466	0.97	0.91 (d = 23)	21	20.3 min	22.4 k



Probability of chemical accuracy (1.6 mHa from FCI) vs d at p = 0.1%; dotted: noiseless.

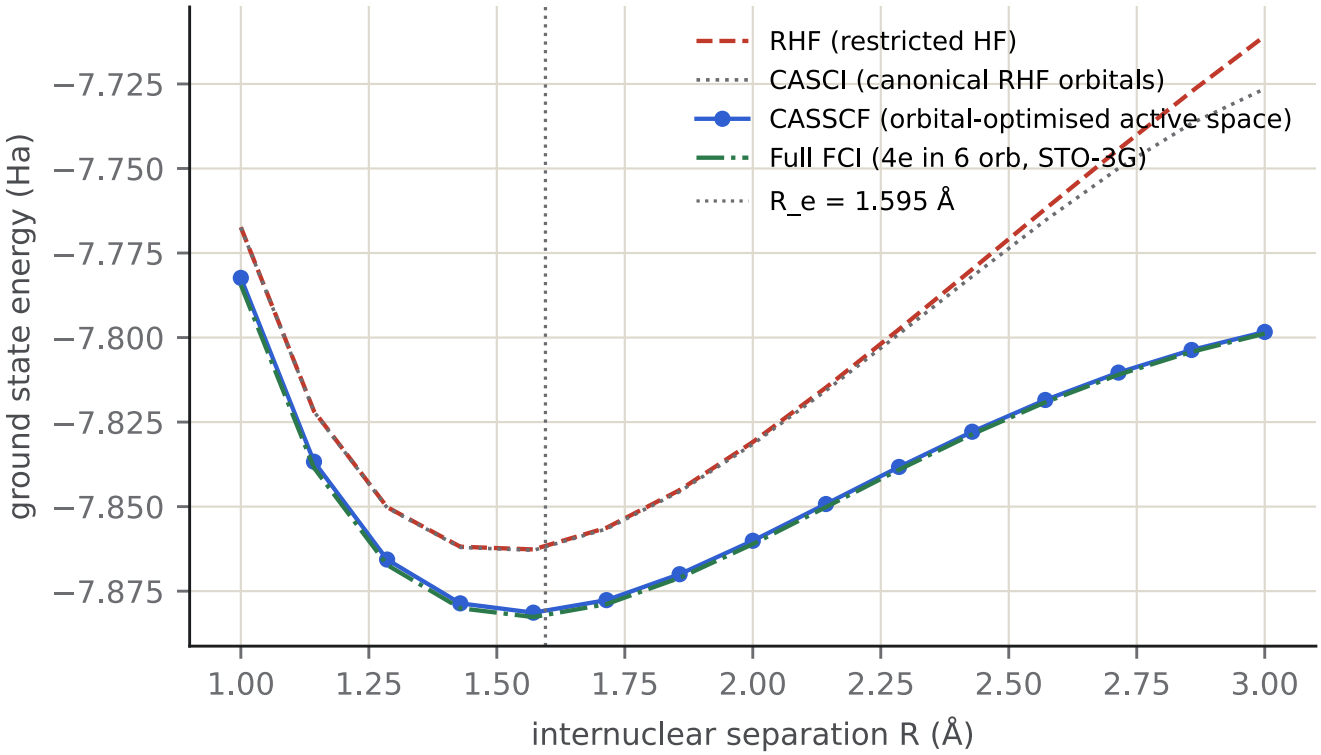
Each run uses ≈ 1.8×10^{7} T gates for H_2, so the magic-state error sets a floor that no distance removes. With cultivated states (2×10^{-9}) that floor is a few percent; with 15-to-1 distilled states (4.5×10^{-8}) most runs fail.



The answer itself: energies returned by 400 individual H_2 runs at several distances (cultivated states, p = 0.1%), as the error from the exact energy; shaded: chemical accuracy; grey: the same circuit without noise.

3.6 LiH

R (Å)	E_RHF (Ha)	E_CASCI (Ha)	E_CASSCF (Ha)	E_FCI (Ha)	Δ(CASSCF - FCI) (mHa)
1.000	-7.767362	-7.767497	-7.782355	-7.784460	+2.105
1.143	-7.821772	-7.821916	-7.836724	-7.838533	+1.809
1.286	-7.850149	-7.850314	-7.865659	-7.867261	+1.602
1.429	-7.861861	-7.862059	-7.878610	-7.880050	+1.440
1.571	-7.862662	-7.862913	-7.881364	-7.882669	+1.304
1.714	-7.856258	-7.856587	-7.877654	-7.878839	+1.185
1.857	-7.845114	-7.845561	-7.869990	-7.871067	+1.077
2.000	-7.830906	-7.831534	-7.860107	-7.861088	+0.981
2.143	-7.814787	-7.815698	-7.849237	-7.850132	+0.895
2.286	-7.797564	-7.798927	-7.838270	-7.839088	+0.817
2.429	-7.779811	-7.781926	-7.827858	-7.828605	+0.747
2.571	-7.761957	-7.765356	-7.818461	-7.819142	+0.681
2.714	-7.744336	-7.749961	-7.810364	-7.810982	+0.618
2.857	-7.727221	-7.736649	-7.803674	-7.804233	+0.559
3.000	-7.710830	-7.726428	-7.798338	-7.798843	+0.505



LiH in STO-3G: restricted Hartree–Fock, CASCI(2, 2) on canonical RHF orbitals, CASSCF(2, 2) with the $1s^2$ core frozen, and full FCI. Dotted: R = 1.595 Å, where phase estimation is run.

With the core frozen, CASSCF(2, 2) stays within 0.5–2.1 mHa of full STO-3G FCI along the curve; CASCI on canonical RHF orbitals is 17–72 mHa above it, and RHF, which cannot dissociate the bond correctly, is 88 mHa above it at R = 3.0 Å. At R = 1.595 Å the active space recovers 19.1 of the 20.4 mHa correlation energy and lies 1.3 mHa above FCI, inside chemical accuracy but using most of it.

Phase estimation of the active-space Hamiltonian (4.3×10^7 T gates per run; 400 shots per configuration, at least 60 per fault stratum):

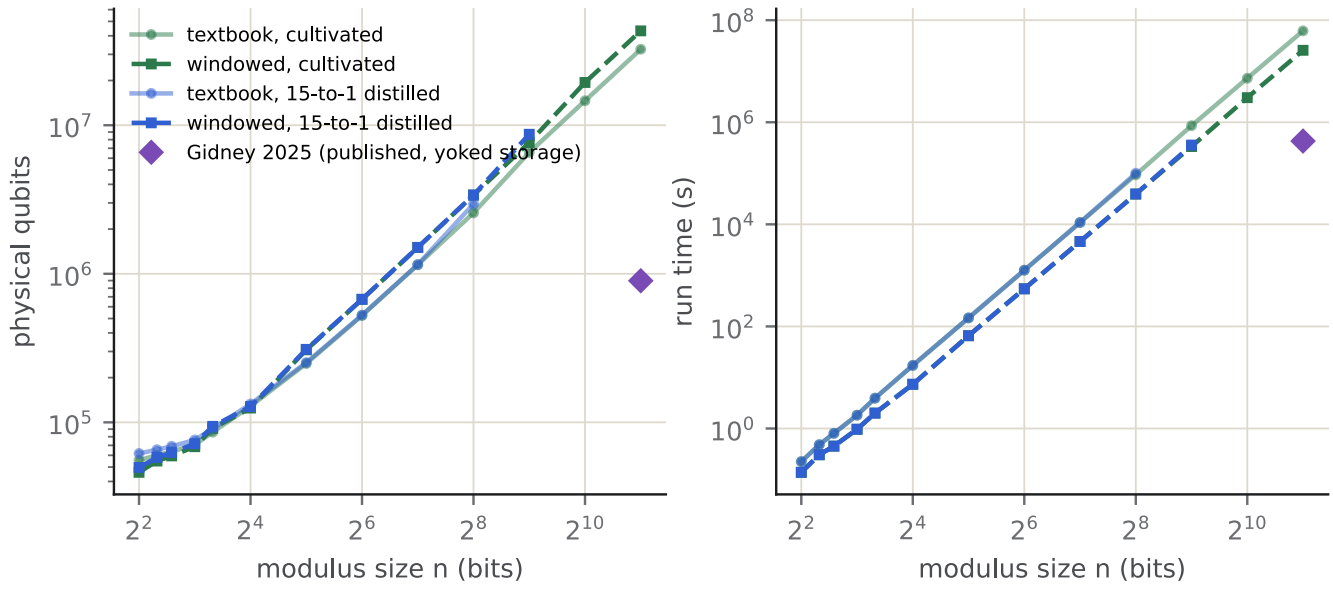
distance	factory	noise p	chem. acc. (CASSCF)	chem. acc. (full FCI)	E[faults]	physical qubits	quantum time
d = 13	cultivated	0.1%	3.0% ± 0.9%	1.5%	41.80	22.1 k	12.4 min
d = 15	cultivated	0.1%	10.8% ± 1.3%	7.2%	5.21	21.3 k	14.3 min
d = 17	cultivated	0.1%	57.7% ± 1.8%	36.6%	0.72	21.1 k	16.2 min
d = 19	cultivated	0.1%	76.5% ± 1.7%	50.8%	0.17	21.5 k	18.1 min
d = 21	cultivated	0.1%	79.4% ± 1.7%	51.0%	9.66e-02	22.4 k	20 min
d = 23	cultivated	0.1%	82.2% ± 1.6%	52.8%	8.78e-02	23.7 k	21.9 min
d = 15	15-to-1 distilled	0.1%	10.0% ± 1.2%	5.1%	7.07	16.1 k	14.3 min
d = 17	15-to-1 distilled	0.1%	25.4% ± 1.9%	14.9%	2.58	16.5 k	16.2 min
d = 19	15-to-1 distilled	0.1%	28.5% ± 1.9%	17.4%	2.03	17.4 k	18.1 min
d = 21	15-to-1 distilled	0.1%	28.1% ± 1.8%	19.0%	1.96	18.7 k	20 min

The noiseless circuit is within chemical accuracy of the active-space energy 88% of the time. With cultivated states the best distance reaches 82% (d = 23); with 15-to-1 distilled states the success saturates near 28%, the magic-state floor of ≈ 2.0 expected faults per run.

3.7 To scale

Operation counts of both RSA-2048 circuits are exact; the schedule structure is fitted on compiled instances (rounds per Toffoli: 5.13 textbook vs. 8.54 windowed; patches busy: 3.18 vs. 2.79). The smallest odd d with at most 0.1 expected faults (p = 0.1%, cultivated magic states):

arithmetic	calibration	logical qubits	Toffolis	d	physical qubits	run time
textbook	separate measure and reset layers	6,150	3.4×10^{11}	35	32.5 M	1.96 years
textbook	one measure-and-reset layer	6,150	3.4×10^{11}	33	29 M	1.85 years
windowed + MBU	separate measure and reset layers	8,199	8.6×10^{10}	35	43.2 M	298 days
windowed + MBU	one measure-and-reset layer	8,199	8.6×10^{10}	33	38.5 M	281 days



Shor scaling from the simulated sizes to RSA-2048 (p = 0.1%). Solid: textbook; dashed: windowed. The diamond is Gidney 2025's published estimate, which stores idle qubits in yoked surface codes.

Windowed arithmetic shortens the RSA-2048 run from 1.96 years to 298 days but needs more logical qubits, so it is not smaller. Both are far from Gidney's 2025 design, whose arithmetic, layout and storage are different in kind; this section measures the error model, not that design.

Published estimates. For published algorithms the logical counts and run time come from the paper and only the error model is ours. Each is compared at its authors' own failure budget and by distance only: their physical-qubit totals depend on layouts (yoked storage, hand-made floor plans) not modelled here. Gidney 2025 is taken one shot at a time (12.07 h, at most 1,409 logical qubits): his budget is per shot, and a failed shot is repeated.

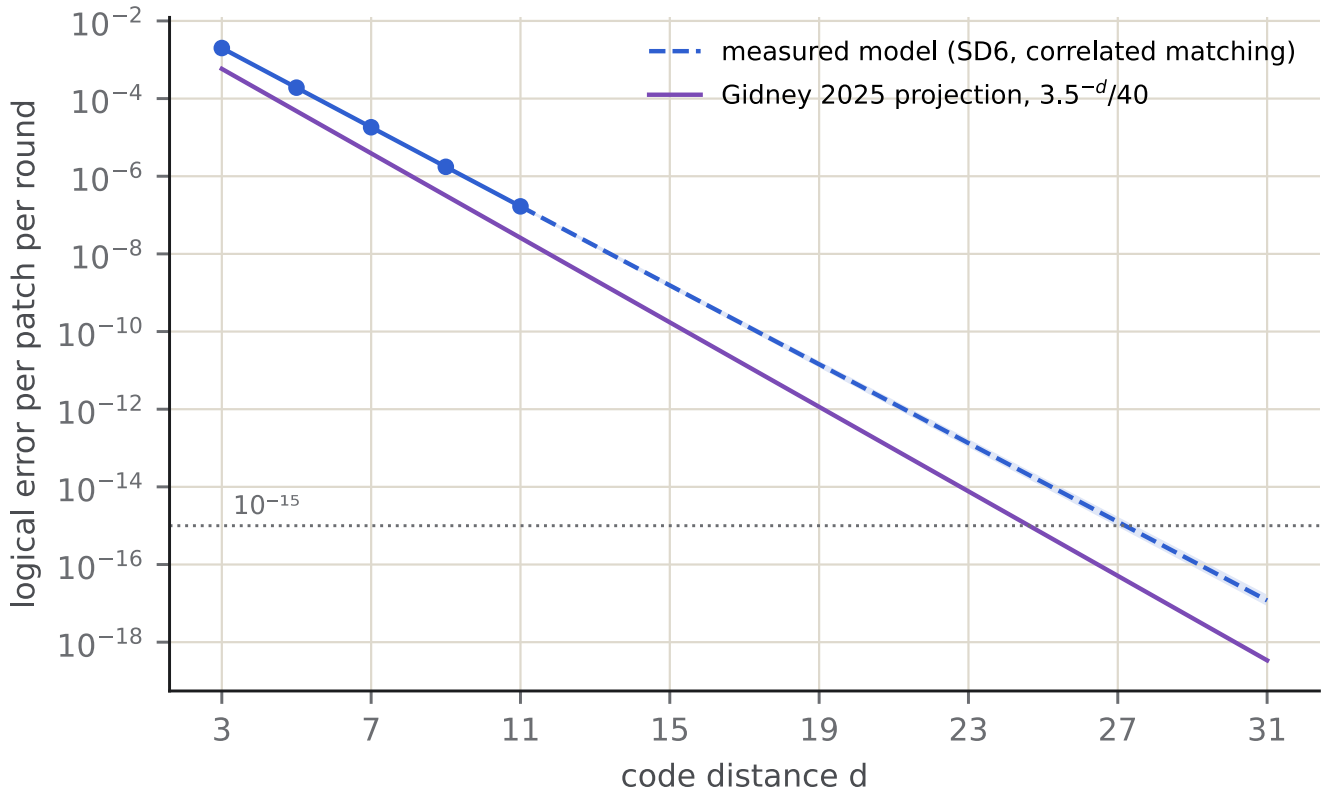
workload	logical qubits	Toffolis	run time	failure budget	published d	d, as calibrated	d, one measure-and-reset layer
RSA-2048, Gidney 2025 (one shot)	1,409	7.1×10^{8}	12.1 h	6.7%	25	29	27
FeMoco (THC), Lee et al. 2021	2,142	5.3×10^{9}	2.45 days	1.0%	31	31	29

3.8 Gidney 2025's distance under the measured model

Gidney chooses d = 25 because his projection of surface-code memory (the legend of his Figure 6: $p_L \approx 3.5^{-d}/40$ per patch per round, from the yoked-surface-code simulations of Gidney, Newman, Brooks and Jones) first reaches 10^{-15} per patch per round there; over a 12-hour shot of up to 1,600 logical qubits that leaves 93.3% of shots free of logical errors. Asked the same question, the measured model answers d = 29 (d = 27–29 within the fit's 1σ band):

d	measured model	1σ band	2σ band	Gidney's projection	ratio
3	2.0×10^{-3}	1.9×10^{-3} – 2.1×10^{-3}	1.8×10^{-3} – 2.2×10^{-3}	5.8×10^{-4}	3.4×
7	1.8×10^{-5}	1.8×10^{-5} – 1.9×10^{-5}	1.7×10^{-5} – 2.0×10^{-5}	3.9×10^{-6}	4.7×
11	1.7×10^{-7}	1.5×10^{-7} – 1.8×10^{-7}	1.4×10^{-7} – 2.0×10^{-7}	2.6×10^{-8}	6.5×
21 *	1.4×10^{-12}	1.1×10^{-12} – 1.7×10^{-12}	9.0×10^{-13} – 2.2×10^{-12}	9.4×10^{-14}	14.8×
23 *	1.3×10^{-13}	1.1×10^{-13} – 1.7×10^{-13}	8.3×10^{-14} – 2.2×10^{-13}	7.7×10^{-15}	17.5×
25 *	1.3×10^{-14}	1.0×10^{-14} – 1.7×10^{-14}	7.7×10^{-15} – 2.2×10^{-14}	6.3×10^{-16}	20.8×
27 *	1.3×10^{-15}	9.4×10^{-16} – 1.7×10^{-15}	7.1×10^{-16} – 2.3×10^{-15}	5.1×10^{-17}	24.7×
29 *	1.2×10^{-16}	9.0×10^{-17} – 1.7×10^{-16}	6.6×10^{-17} – 2.3×10^{-16}	4.2×10^{-18}	29.4×
31 *	1.2×10^{-17}	8.5×10^{-18} – 1.7×10^{-17}	6.1×10^{-18} – 2.4×10^{-17}	3.4×10^{-19}	35.0×

* extrapolated beyond the measured distances (d ≤ 11). Bands are the weighted least-squares fits' parameter covariance, both bases moved together.



Logical error per patch per round of an idle patch at p = 0.1%: the measured model (points measured, line fitted, dashed beyond the largest measured distance, shaded 1σ) and Gidney's projection. Dotted: the 10^{-15} he designs to.

The gap is not only extrapolation. At the measured distances SD6 memories decoded by correlated matching already fail 3.4–6.5× more often than the projection, and the fitted suppression per two units of distance (Λ = 10.1 and 10.9 for the two bases) is below its 12.25, so the ratio grows to 21× at d = 25. The two are fits to different simulations; Section 3.9 separates the circuit from the noise model.

Running one shot of his algorithm through the estimator of this report:

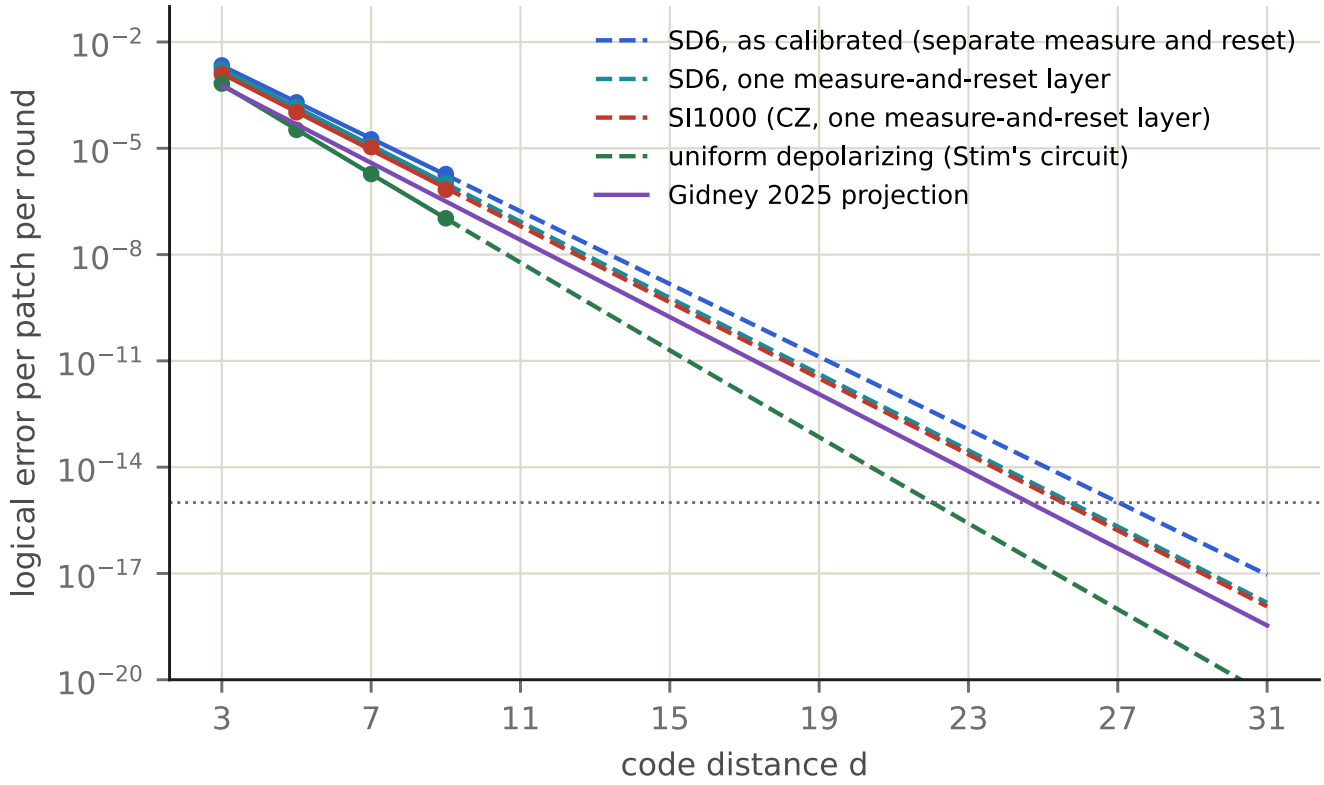
varied	scenario	d	Δd
baseline	one shot at his budget (E[faults] ≤ 0.069)	29	0
Idle fit	rates at the −2σ edge of the fit	27	-2
Idle fit	rates at the −1σ edge of the fit	27	-2
Idle fit	rates at the +1σ edge of the fit	29	+0
Idle fit	rates at the +2σ edge of the fit	29	+0
Extrapolation	Λ × 1.15 beyond the measured distances	27	-2
Magic states	ε_CCZ × 0.1	29	+0
Magic states	ε_CCZ × 10	29	+0
Budget	90% of shots succeed (E[faults] ≤ 0.1)	27	-2
Budget	99% of shots succeed (E[faults] ≤ 0.01)	29	+0
his model	Gidney's projection in place of the measured model	25	-4

At his budget the measured model needs d = 29 (d = 27 if 90% of shots may succeed). The fit's uncertainty and a 15% steeper extrapolation move it by at most two; the magic states do not move it at all, since idling is 99% of the expected faults. Given his projection instead, the same estimator returns his d = 25. Lee et al.'s FeMoco estimate assumes a suppression of 10 per two units of distance, close to the measured 10.1–10.9, and under the measured model it comes out at their d = 31. None of this measures Gidney's design, which would need his layout, its yoked storage and his decoder simulated directly.

3.9 Which memory model?

The measured model is one circuit under one noise model. To see how much the distance depends on that choice, the same memory experiments (d and 3d rounds, both bases, correlated matching, per-round rates from the ratio of the two) were run at p = 0.1% on three more: the same SD6 circuit with each round's measurement and reset of the ancillas in one layer, as hardware-style circuits have them; SI1000, the superconducting-inspired model of the yoked-surface-code paper's simulations (Table 3 of arXiv:2312.04522: CZ p, single-qubit gates and idling p/10, reset 2p, measurement 5p, qubits waiting for a measurement or reset 2p), on the circuit rewritten over CZ with one measure-and-reset layer; and Stim's generated memory with every parameter at p, one reading of the "uniform depolarizing" that labels Gidney 2025's Figure 6, which leaves qubits idle during gate layers noiseless. The noise is applied layer by layer by ftalgo.calib.circuit_noise, which given SD6 reproduces stabilizer-qec's own circuit's error model exactly.

memory model (p = 0.1%)	Λ (Z, X)	largest d	rate / projection there	d: worse basis ≤ 10^{-15}	d: either basis ≤ 10^{-15}	d: one shot at his budget
SD6, as calibrated (separate measure and reset)	10.8, 10.5	9	5.9×	27	29	27
SD6, one measure-and-reset layer	12.0, 11.8	9	3.2×	27	27	27
SI1000 (CZ, one measure-and-reset layer)	11.6, 12.3	9	2.1×	27	27	27
uniform depolarizing (Stim's circuit)	20.4, 15.9	9	0.3×	23	23	23
Gidney 2025 projection	12.25	–	1×	25	25	25



Logical error per patch per round (either basis) at p = 0.1% under each memory model: points measured to d = 9, lines fitted, dashed beyond; and Gidney's projection. Dotted: 10^{-15}.

The distance that reaches 10^{-15} per patch-round runs from d = 23 to d = 29 across the four models, and Gidney's d = 25 lies inside that range. SI1000 has the projection's slope here (Λ = 11.6–12.3) but a 2.1× higher rate at d = 9; Stim's circuit is below the projection. The calibration's circuit is the most pessimistic: it depolarizes idle qubits in every layer and gives data qubits two idle layers a round, one for the ancillas' measurement and one for their reset. Merging the two lowers the d = 9 rates 1.7–2.0× and raises Λ to ≈ 12. Every simulated algorithm run in this report rests on that circuit and leans pessimistic accordingly. (The SD6 row fits the calibration's memories like the others, over d ≤ 9; with all its memory lengths and d = 11 the calibrated model needs d = 29.)

Recalibrated with one measure-and-reset layer. Every calibration experiment was rerun on stabilizer-qec's circuits with each reset layer folded into the measurement layer before it (ftalgo.calib.circuit_noise.merge_measure_reset_layers: every noise channel kept except the reset layer's idle depolarizing; on memories it is the SD6 one-layer model above exactly). Idling then has $\Lambda = 11.6$–12.5 at $p = 0.1\%$, reaches 10^{-15} per patch-round at $d = 27$ ($d = 27$ at either 1σ edge), and one shot of Gidney's algorithm needs $d = 27$ at his budget; textbook RSA-2048 needs $d = 33$ (29 M physical qubits). Whole surgery experiments predicted from its parts are 0.87–1.14× the measured failure rate (median 1.02×), as for the original calibration. Section 3.7's tables give both calibrations.

The simulated algorithm runs were repeated on the recalibrated machine as well (16 workloads with cultivated magic states, plus the injected and 15-to-1 factories; data/results_mr/, run in parallel on GitHub Actions by tools/grid.py, which splits long configurations into shot chunks and pools them). The distance at which a run first reaches 90% of its noiseless success drops by one step for 9 of them and stays the same for 7; none needs more. Where the magic states set the ceiling (injected states, H_2 with 15-to-1 distillation), the ceiling is unchanged within error.

workload	p	noiseless	d for 90%, as calibrated	one measure-and-reset layer
Shor, textbook, N = 15	0.1%	1.00	13	11
Shor, textbook, N = 15	0.2%	1.00	19	17
Shor, textbook, N = 21	0.1%	0.79	13	13
Shor, textbook, N = 21	0.2%	0.79	21	19
Shor, textbook, N = 35	0.1%	0.79	15	13
Shor, textbook, N = 35	0.2%	0.79	21	21
Shor, textbook, N = 77	0.1%	0.77	15	15
Shor, textbook, N = 77	0.2%	0.77	23	21
Shor, textbook, N = 143	0.1%	0.77	15	15
Shor, textbook, N = 143	0.2%	0.77	23	21
Shor, windowed + MBU, N = 15	0.1%	1.00	11	11
Shor, windowed + MBU, N = 21	0.1%	0.79	13	13
Shor, windowed + MBU, N = 35	0.1%	0.78	13	13
H2	0.1%	0.92	19	17
HeH+	0.1%	0.97	21	19
LiH, CAS(2, 2)	0.1%	0.88	23	21

4 Validation

- ftsim: dense = sparse on random circuits; 100 random circuits equal Qiskit's statevector to 10^{-10}; noisy distributions equal Aer's exact density matrices (χ^2, plain and stratified); WebAssembly = native.
- Arithmetic, textbook and windowed, verified on every input by an independent bit-level simulator, including the measurement-based uncomputation and its feed-forward corrections; Shor's noiseless outcomes match the exact distribution; synthesized rotations verified as matrices; QPE matches its exact Trotterized distribution.
- Scaling: the Toffoli and qubit counts used at RSA-2048 equal the compiled circuits' for every simulated N; the estimator, given Gidney 2025's projection and budget, returns his d = 25.
- Chemistry: integrals, HF and FCI equal PySCF to 10^{-8} Ha; Szabo and Ostlund's H_2 reproduced; tapering preserves the spectrum; the LiH active-space Hamiltonian's ground energy equals PySCF's CASSCF(2, 2) to 10^{-9} Ha (and CASCI on RHF orbitals to 10^{-11} Ha).
- Calibration: Stim + PyMatching cross-check; reference-qubit experiment against plain memories; composition test. The sampling-and-decoding runner is checked against PyMatching's correlated decoder on the same samples (stabilizer-qec 2.x, like PyMatching, decodes with correlations only when asked at decode time as well; the calibration was run with 1.2).
- Noise models: SD6 applied layer by layer reproduces stabilizer-qec's circuit's error model exactly; the CZ rewrite is deterministic with graphlike distance d; SI1000's rates are Table 3's.

5 Limitations

- Composition is accurate to about 10% (Section 3.2), measured only for d ≤ 7.
- Logical channels are measured to d = 11 (idling) and d = 9 (surgery) at p = 0.1%. Most algorithm results, and every RSA-2048 and FeMoco number, lie beyond that, along exponential fits; they are marked.
- Magic-state error rates and footprints, the H and S operation model and the timing constants are cited inputs, not measured here.
- Published algorithms enter only through their gate counts and run times; their layouts (Gidney 2025's yoked storage and parallel lookups, Lee et al.'s floor plan) are not modelled, so only distances are compared.
- The LiH problem is a 2-qubit active space; chemistry at scale is represented only by FeMoco's published counts.
- The machine is calibrated on stabilizer-qec's SD6 memory and surgery circuits, which measure and reset ancillas in separate layers (the pessimistic end of four circuit-level models, Section 3.9), and on the same circuits with one measure-and-reset layer; the figures and per-run tables use the first, and Section 3.9 compares the two. Biased noise and the codes tailored to it (XZZX) are not modelled.

6 Reproduction

```
tools/build.sh                     # the engine (Rust → Python module)
python tools/calibrate.py          # circuit-level experiments (resumable)
python tools/fit_model.py          # data/calibration/model.json
python tools/compose_check.py; python tools/xcheck.py
python tools/noise_models.py; python tools/noise_models.py --summary
python tools/calibrate.py --mr; python tools/fit_model.py --mr; python tools/compose_check.py --mr
python tools/run_shor.py; python tools/run_shor_modern.py
python tools/run_qpe.py; python tools/run_lih.py
python tools/grid.py plan|run|merge   # the same runs on model_mr.json (or Actions → Algorithm grid)
python tools/scale.py; python tools/sensitivity.py
python tools/site_data.py; python tools/report.py
```